



MINING DEALER BEST PRACTICE

D11R/T Converter Removal & Installation Tool

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The lifting devices used for the removal and installation of components in the equipment must provide protection to the personnel and machine. The technicians in a shop don't always have a specific tool and use some general methods for the removal of components. The Converter for the D11R is an example of this. Such a component is removed by using two ratchet pullers, link brackets, and slings (Figure 1).

This Best Practice describes the design of a tool to remove the D11R/T torque converter in a more secure and fast for technicians.

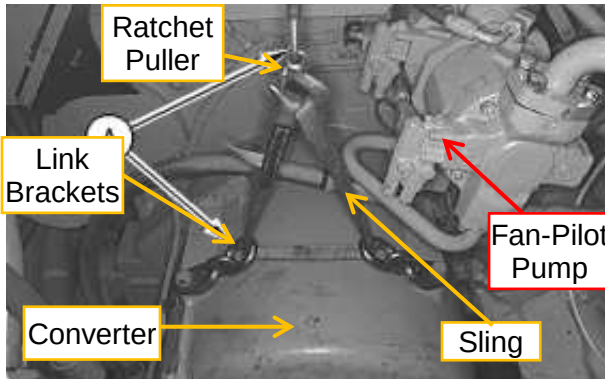


Figure 1. Converter removal according to the Disassembly and Assembly Manual (SENR 1782-08)

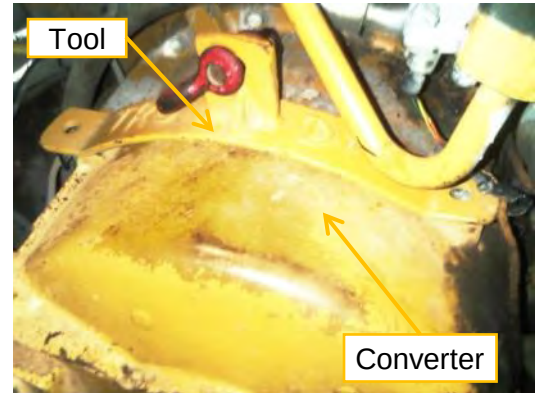


Figure 2. Converter removal according to proposal

2.0 Best Practice Description

The tool consists of a curved plate to be adjusted to the converter geometry. The dimensions can be found attached to the drawings (See Figure 4).

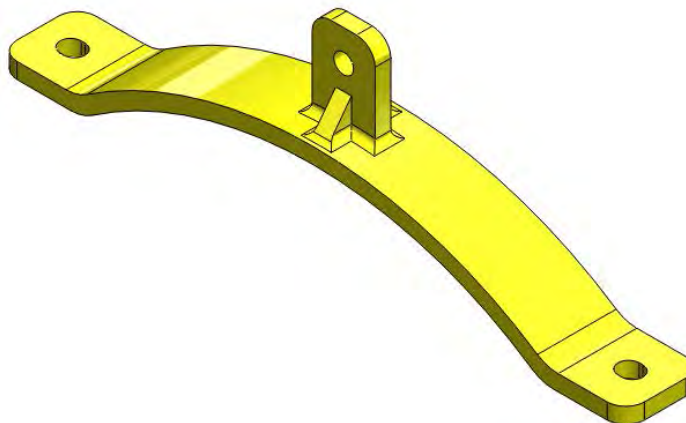


Figure 2. Converter Removal & Installation Tool

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3.0 Implementation Steps

Steps for removal

1. Refer to the SENR 1782-08 Manual (D11R Torque Converter Removal) or the Manual KENR 5620-00 (D11T Torque Converter Removal), and follow the steps before installing the tool.
2. Locate the tool to the converter using two 5/8" bolts – 11 x 1.75" UNC (grade 8) with their washers. The minimum torque value for the bolts is 30 N·m.
3. The converter is removed with a lifting device with a minimum lifting capacity of 550 kg. The weight of the torque converter is approximately 366 kg.

Steps for the Installation

1. Install converter tool with bolts and washers listed above (minimum torque value 30 N·m)
2. Refer to the Installation Manual SENR 1782-08 (D11R Torque Converter Installation) or to the Manual KENR 5620-00 (D11T Torque Converter Installation).

The tool can be used in the followings converters (D11R and D11T series)

Table 1. Converter that apply		
Model	Part Number	Description
D11R	121-0030 ¹	Torque Converter Group
	129-4041	Torque Converter Group
D11T	343-9846	Torque Converter & Housing Group
	375-2685	Torque Converter & Housing Group
	177-9790	Torque Converter Group

¹ Some D11T models use this converter

4.0 Benefits

The tool has the following advantages:

- Reduced assembly time.
- A faster converter removal process when reducing the possibilities of contacting the fan pilot pump, not using slings (See Figure 1).
- Lower cost of the tool. See Comparative table 2 and 3.

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Table 2. Required Tools (SIS Cat)				
Tool	Part Number	Item Weight (Kg)	Qty	Price (USD)
Ratchet Puller	8S-9906	3.1745	2	\$ 640.25
Link Bracket	138-7574	0.975	3	\$ 198.41
5/8 – 11 x 1.50 inch Bolt	0S-1566	0.0907	2	\$ 3.34
Nylon Lifting Sling	1U-8234	0.1769	2	\$ 39.33
Total		9.6278		\$ 881.33

Table 3. Required Tools (Proposal)				
Tool	Part Number	Item Weight (Kg)	Qty	Price (USD)
Lifting Tool	138-7574	3.68	1	\$ 437.91
Shackle (2 Ton)	189-0408	0.3175	1	\$ 21.27
5/8 – 11 x 1.75 inch Bolt	0S-1589	0.0998	2	\$ 3.93
5/8 inch Washer	5P-8247	0.0136	2	\$ 2.30
Total		3.9975		\$ 463.10

5.0 Resources Required

- 2 5/8 Bolts - 11 x 1.25 inch UNC (P/N: 0S-1592).
- 2 5/8 Washers (P/N: 5P-8247).
- Material for making the tool (See Figure 4 for dimensions)
- Technicians.

6.0 Supporting Attachments / References

- SIS document: SENR 1782-08 (Torque Converter Remove – SMCS: 3101-011)
- SIS document: SENR 1782-08 (Torque Converter Install – SMCS: 3101-012)
- SIS document: KENR 5620-00 (Torque Converter Remove – SMCS: 3101-011)
- SIS document: KENR 5620-00 (Torque Converter Install – SMCS: 3101-012)
- D11R Converter R&I Tool Drawing
- Finite Element Analysis for R&I Tool

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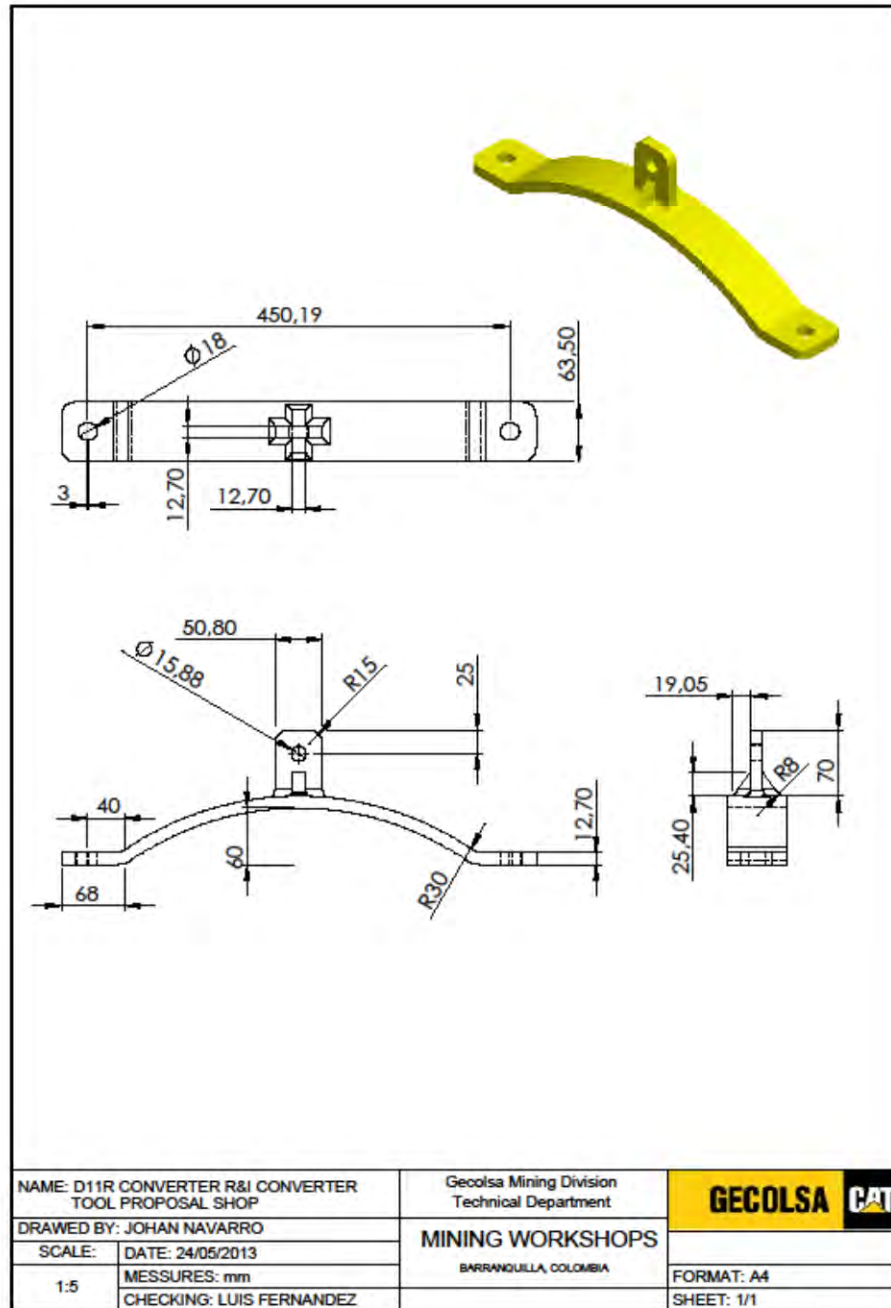


Figure 4. Tool Drawing.

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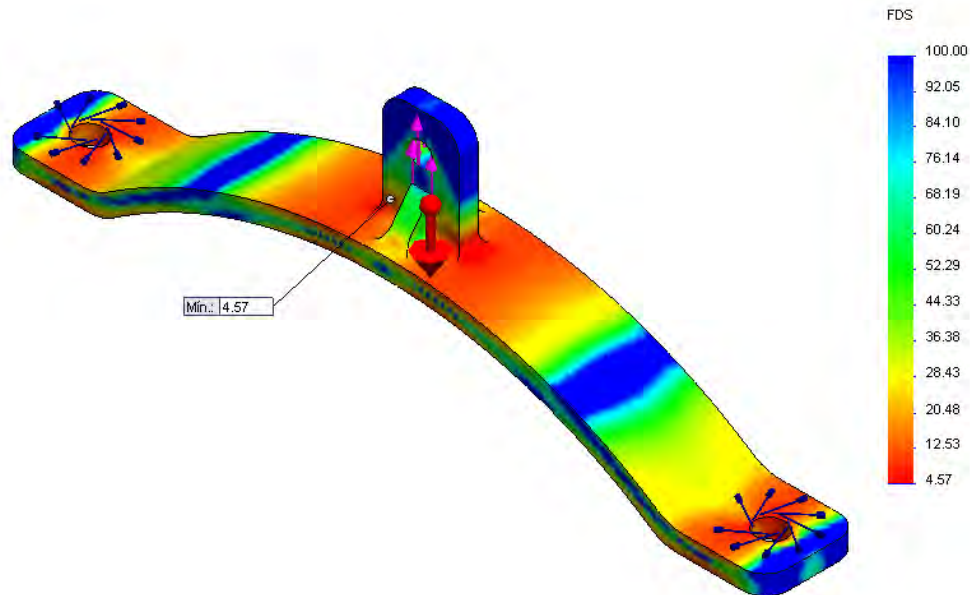


Figure 5. Safety Factor Analysis

Physical Properties		
Material	AISI 1045 Steel, cold drawn	
Mass	3.675 Kg	
Density	7850 Kg/m ³	
Yield Strength	530 MPa	
Tensile Strength	625 MPa	
Shear Modulus	80 Gpa	
Study Properties		
Analysis type	Static	
Mesh type	Solid mesh	
Mesh Quality	Second order tetrahedral elements	
Model type	Lineal elastic isotropic	
Failure criteria	Von Mises Stress Maximum	
Fixtures		
Type	Fixtures details	
Foundation Bolt	Quantity	2
	Nominal Diameter	5/8 inch
	Pre-load (Torque)	30 N·m
Virtual Wall	Entities	2 faces (below faces)
	Wall type	Bonded
	Friction coefficient	0.15
Loads		
Gravity	9.81 m/s ²	
Vertical Force	4000 N (400 kg approx.)	
Study Results		
Von Mises Stress Maximum	115.9 Mpa	
Minimum Safety Factor	4.57	

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Based on finite element analysis and considering the minimum safety factor given by the *ASME B30.20-2003* standard: *Below-the-Hook lifting device*, the tool loading capacity is 600 Kg.

7.0 Related Best Practices

8.0 Acknowledgements

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